

Coding Assignment 1

ECE 414/517 — Dynamic Programming on RiverSwim

Setup. The supplied six-state model starts at s_0 (array index 0). Action 0 is LEFT and action 1 is RIGHT. LEFT at s_0 pays 0.005; RIGHT at s_5 pays 1. Values are expected infinite-horizon discounted returns. Value iteration and the policy-evaluation step use synchronous updates, stopping when the maximum absolute change in the value vector is at most 10^{-10} . Policy iteration stops when its policy is unchanged. All supplied iteration limits are retained. Greedy ties select the smallest action index.

(a) Largest grid discount selecting LEFT. I evaluated every discount in $\{0.00, 0.01, \dots, 0.99\}$ for every current strength. The initial action is LEFT through the second column of the table and RIGHT from the third column onward on this grid. These are *grid thresholds*, not exact continuous switching points. The corresponding switching intervals are $[0.61, 0.62]$, $[0.71, 0.72]$, and $[0.92, 0.93]$ for weak, medium, and strong currents.

Current	Last LEFT γ	First RIGHT γ	$V^*(s_0)$ at .99	$V^*(s_0)$ at .999
Weak	0.61	0.62	73.471119	843.931942
Medium	0.71	0.72	47.388319	587.773891
Strong	0.92	0.93	3.223457	46.106373

The .999 values come from the graduate-extension grid. At both .99 and .999, the optimal policy for all three currents is $[1, 1, 1, 1, 1, 1]$: choose RIGHT in every state.

(b) Policy iteration versus value iteration. Both methods converge for every tested setting and return identical six-state policy arrays in all 300 required-grid cases and all 150 extension cases. The maximum absolute difference between their value vectors, over all states and discounts, is shown below.

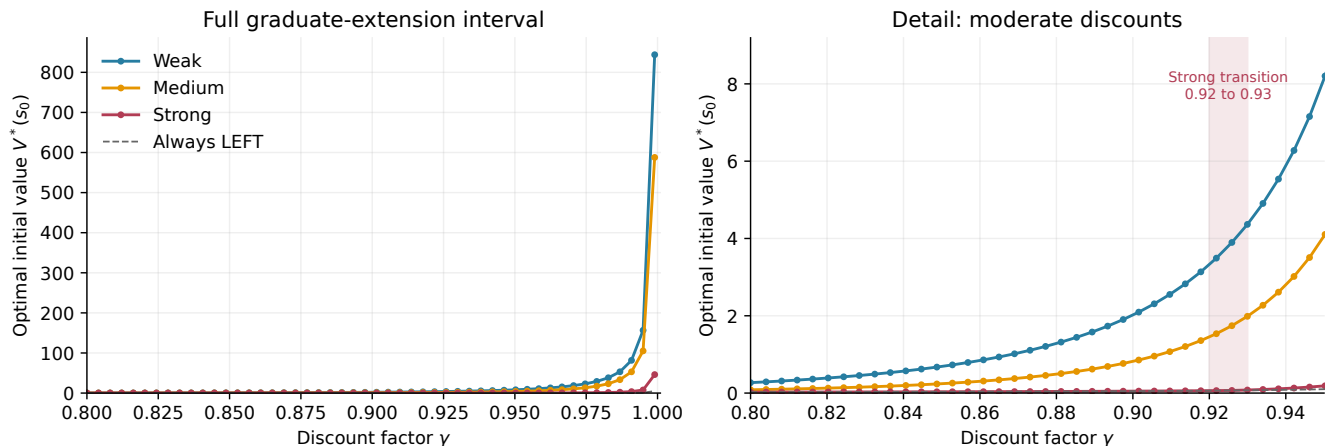
Current	Required grid	Extension grid
Weak	7.82×10^{-14}	1.07×10^{-14}
Medium	1.44×10^{-12}	2.50×10^{-13}
Strong	8.12×10^{-11}	9.31×10^{-11}

The overall difference is 9.31×10^{-11} . As an independent numerical check, each returned policy was evaluated with $(I - \gamma P^\pi)^{-1} r^\pi$ using a linear-system solve and confirmed greedy with respect to those values. Across both methods and both grids, the largest optimal Bellman residual is 9.99×10^{-11} and the largest difference from direct evaluation is 9.99×10^{-8} . The stopping tolerance bounds successive updates, not the exact value error: the contraction bound is approximately $10^{-10}/(1 - \gamma)$, or 10^{-7} at $\gamma = .999$.

(c) Effect of stronger current. The RIGHT dynamics list backward, stay, and forward probabilities: weak (.05, .60, .35), medium (.10, .65, .25), and strong (.15, .75, .10). The interior expected displacement under RIGHT therefore decreases from +.30 to +.15 to $-.05$ states per step. Reaching and remaining near the larger right-boundary reward becomes harder, so a longer effective horizon is needed to justify leaving the small guaranteed reward on the left. The largest LEFT discount increases by 0.10 from weak to medium and by another 0.21 from medium to strong. At a fixed sampled discount, $V^*(s_0)$ never increases with current strength: it is weak $\geq$ medium $\geq$ strong on both grids. At .99 the strong-current value is 95.6% below the weak-current value.

Graduate extension and reproducibility

ECE 517 extension. I evaluated 50 evenly spaced discounts using `np.linspace(0.80, 0.999, 50)` for each current strength. This includes both endpoints and exceeds the required 20 points. The figure plots the initial optimal value from value iteration; policy iteration gives the same curves to numerical precision.



Each colored marker is one of the 50 tested discounts. The right panel enlarges the moderate-discount range; the shaded band brackets the strong-current transition on the required 0.01 grid. The dashed curve is the value of staying at s_0 and choosing LEFT forever.

For weak and medium currents, all 50 extension discounts choose RIGHT at s_0 . For the strong current, 31 choose LEFT and 19 choose RIGHT. Whenever LEFT is optimal at the initial state, the agent remains there and receives $V^*(s_0) = 0.005/(1 - \gamma)$. For example, all three currents have value 0.0125 at $\gamma = .60$; the stronger currents separate only when swimming right becomes worthwhile.

As γ approaches 1, future rewards retain more weight and the effective horizon $1/(1 - \gamma)$ grows from 5 at .80 to 1,000 at .999. Repeated rewards at the right boundary then compensate for travel delays, even against the strong current. All three values rise sharply, while the weak current still yields the highest return because it reaches and occupies the right boundary more easily. The strong-current value remains 94.5% below the weak-current value at .999. The LEFT baseline also diverges as γ approaches 1, so rising value alone does not imply a policy change; the relative action values determine it.